VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI -590 014

Mini-Project Report on

"A Web-based Blood Bank System for Managing Records of Donors and Recipients"

Submitted in partial fulfillment of the requirement for the award of degree of

BACHELOR OF ENGINEERING
IN
COMPUTER SCIENCE AND ENGINEERING

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2025-2026

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S.J.M INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(Affiliated to Visvesvaraya Technological University, Belagavi, Recognized by AICTE, New Delhi and Approved by Government of Karnataka)

NAAC Accredited with 'B^{++}' Grade

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CERTIFICATE

This is to certify that the Mini-Project work entitled **" A Web-based Blood Bank System for Managing Records of Donors and Recipients"** is a bonafide work carried out by **SOMESHA K** bearing **USN: 4SM23CS093, UMMAR FARUKH** bearing **USN: 4SM23CS109, VIKAS** bearing **USN: 4SM23CS117, VIKAS R RATHOD** bearing **USN: 4SM23CS118** in partial fulfillment for the award of degree of **Bachelor of Engineering in Computer Science & Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year 2025–2026. It is certified that all corrections/suggestions indicated during Internal Assessment have been incorporated in the report deposited in the departmental library. The Mini-Project report has been approved as it satisfies the academic requirements in respect of Mini-Project work prescribed for the Bachelor Degree.

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ABSTRACT

The Web-based Blood Bank System is a project developed to efficiently manage and store records of blood donors and recipients. The project was developed using **PHP** for the front-end and server-side scripting, and **MySQL** as the database for storing donor and recipient information. The main objective of this project is to make blood donation and request processes faster, easier, and more organized. It allows users to register as donors, check available blood types, and request blood online. The administrator can manage donor details, update blood stock, and handle blood requests through the system. By implementing a web-based platform, the project reduces manual work and paperwork, ensuring accuracy and real-time accessibility of information. Overall, this project was developed to improve the efficiency, security, and reliability of blood donation management through a user-friendly digital solution.

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CHAPTER 1

INTRODUCTION

A Web-based Blood Bank System is designed to securely manage donor and recipient records online.It helps hospitals and blood banks track blood availability in real time.The system reduces manual work by automating data entry and record management.Users can easily search for required blood groups and make requests quickly.Real-time updates help during emergencies by speeding up the matching process.The platform ensures data accuracy, transparency, and fast access from anywhere.Overall, it improves the efficiency and reliability of blood bank operations

1.1 Objectives:

- Efficient Management of Records – Maintain accurate records of donors and recipients, including personal details, blood type, donation history, and medical requirements.
- Quick Access to Information – Enable fast retrieval of donor and recipient data for emergencies and reporting purposes.
- Manage blood inventory efficiently and track availability.
- Reduce manual errors in record-keeping.
- Ensure required blood types are available when needed, especially during emergencies or critical situations.

1.2 Problem Statement:

Blood banks play a critical role in saving lives, but traditional methods of managing donor and recipient records are often inefficient and prone to errors. Most blood banks rely on manual record-keeping, which can lead to misplaced files, delayed information retrieval, and difficulty in tracking blood inventory. This can result in critical delays during emergencies, improper stock management, and sometimes the unavailability of required blood types. There is a need for a reliable, organized, and automated system that can efficiently manage donor and recipient information, track blood inventory in real-time, reduce errors, and ensure timely availability of blood. A web-based solution can address these issues by providing a centralized platform accessible to authorized personnel, improving efficiency, accuracy, and overall management of the blood bank operations.

1.3 Existing System:

In the existing system, blood banks generally use manual methods or basic software to manage donor and recipient records. Most of the data is stored in paper files or simple spreadsheets, which makes searching, updating, and tracking information time-consuming and prone to errors. Blood inventory is often not monitored in real-time, leading to shortages or wastage due to expired units. Communication with donors and recipients is limited, and notifying them about donation schedules or blood availability is inefficient. Overall, the existing system lacks automation, reliability, and easy accessibility, which can cause delays in emergency situations and reduce the overall efficiency of blood bank operations.

Disadvantages:

- Records are maintained manually, leading to **high chances of errors** and data inconsistency.
- Searching for donor details or blood availability is **time-consuming** and inefficient.
- **No real-time updates**, causing delays in emergencies when quick decisions are needed.
- Physical records are easily **misplaced, damaged, or lost** over time.
- Communication between hospitals, donors, and recipients is **slow and unorganized**.
- Manual verification of donor eligibility is **tedious and unreliable**.
- Maintaining and updating large volumes of data becomes **difficult and unmanageable**.

1.4 Proposed System:

The proposed web-based blood bank system provides a centralized and automated platform for managing donor and recipient information. Users can easily register, update profiles, and check the availability of required blood groups through an online interface. Hospitals and administrators can track blood stocks, process requests, and manage donor eligibility efficiently. The system ensures accuracy by reducing manual paperwork and minimizing data entry errors. Real-time data access enables quicker decision-making during emergency situations. A secure backend database stores all records with proper authentication and authorization. The platform improves coordination between donors, recipients, and hospitals. Overall, the system enhances reliability, speed, and transparency in blood bank operations.

Advantages:

- Provides **real-time access** to blood availability and donor information.
- Reduces **manual errors** through automated data entry and processing.
- Ensures **faster communication** between donors, recipients, and hospitals.
- Offers a **user-friendly interface** for easy registration and record management.
- Improves **data accuracy and reliability** with a centralized database.
- Enhances **emergency response** by quickly matching donors with recipients.
- Strengthens **security** with authentication and controlled access.
- Allows easy **updating, tracking, and reporting** of blood inventory.

CHAPTER 2

LITERATURE SURVEY

1. **Shah, A., Shah, D., Shah, D., Chordiya, D., Doshi, N., & Dwivedi, R. (2022). Blood Bank Management and Inventory Control Database Management System. Procedia Computer Science, 198, 404–409.**

This paper presents a detailed approach for an efficient blood bank database management system. The database is the single most useful setting for caching data, and it is also an ideal tool for contriving, managing, updating, and modifying data from different angles. The benefits of a well-structured blood bank database are limitless and yield the benefits of improving efficiency and saving time. Here, our motive is centred on this area. India faces a shortage when it comes to the amount of blood donated. The gap in demand and supply in widened due to mismanagement and inefficient databases. We have modelled a well-organized database to try and reduce this gap. Alongside, we have developed an application that reminds donors when they become eligible again, gives locations of nearby blood donation camps, makes requesting blood easier for blood recipients etc. as well as promoting a healthy community. IOT is used for interlinking the application to the server as well as for inter-application communication. With the help of IOT this collection and exchange of data becomes more efficient.

2. **Ali, A., Jahan, I., Islam, A., & Parvez, S. (2015). Blood donation management system. American Journal of Engineering Research, 4(6), 123–136.**

This paper is focused on Blood Donation Management System which is a web application with supporting mobile application aimed to serve as a communication tool between patients (who need blood) and blood donor. To become members of the system, donors need to create their profiles by providing fundamental information like name, blood group, email address, password, and exact location from "Google Map". In order to find out the exact location of a donor, Google Map is integrated with this application. The mobile application always updates the location of a donor. As a result, the system can automatically find a registered donor wherever he/she goes. Visitors can search blood donors from the home page by blood group and the place where blood is needed. The system will show the available donors along with their phone number, email address and mailing address through arranging them by nearest place and blood donation expire date. Visitors can send message to all donors through email but a member can send message using email and mobile phone. An appointment will be created only whenever a donor confirms that he/she will donate blood. Then the system will alert the donor before 12 hours of donation. Blood donors can also be searched from the mobile application, but this is only accessible for registered members.

3. **Izadeen, G. Y., Abdulazeez, A. M., Zeebaree, D. Q., Hasan, D. A., & Ahmed, F. Y. (2021). Data Integration Using Data Mining and SMS Reminder for Automation of Blood Donation. IEEE I2CACIS, 299–304.**

Blood is one of the most vital and essential elements in human existence. When the population increases, so do the request for blood. People who need blood in an emergency are unable to provide it promptly. This paper suggests an effective method of contacting donors, that can be useful in an emergency. When a person requires blood, they request it through a website or mobile device; the request is then routed to the person who meets the matching blood type. the application is then sent an SMS to the donor to approve it, after accepting the request, the application will inform the requestor about it and he/she will get the donor phone number by using an MCU ESP8266 and a SIM800L. The privacy of the person must be protected in the current environment. The donor will be deleted from the reception of notification for the next three months after the blood donation is done. User name, password, and phone number are used to check registered accounts.

4. **Kayode, A. A., Adeniyi, A. E., Ogundokun, R. O., & Ochigbo, S. A. (2019). An Android based blood bank information retrieval system. Journal of Blood Medicine, 10, 119.**

Background:

Blood Bank record keeping has been carried out manually over the past decades using paper file management system which is slow for information retrieval and processing and also prone to errors in an emergency situation. **Materials and methods:** This research work solves the above-mentioned problem with the development of both web-based and Android-based blood bank information retrieval system. The web application is used by various blood banks system administrators to update their available blood inventory information and the mobile application which has the mobile search engine is used to search for blood supplies from the registered blood banks. **Results and conclusion:** The system also has a feature that allows registered blood banks to send a notification to registered blood donors on the application requesting for blood donation.

CHAPTER 3

SOFTWARE AND HARDWARE REQUIREMENTS

3.1 SOFTWARE REQUIREMENT

• Operating System: Windows 7/8/10 or Linux

• Front End: PHP, HTML, CSS

• Database: MySQL

• Server: XAMPP

• Design Tool: PHP

3.2 HARDWARE REQUIREMENT

• Processor: x86 compatible processor with 1.7 GHz Clock Speed

• RAM: 512 MB or greater

• Hard Disk: 20 GB or grater

• Monitor: VGA/SVGA

• Keyboard: 104 keys standard

• Mouse: 2/3 button, Optical/Mechanical.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

In the Web-based Blood Bank System for managing donor and recipient records, the architecture follows a client-server model, where users access the system via web or mobile apps, and the server handles business logic, data processing, a secure management of blood inventory.

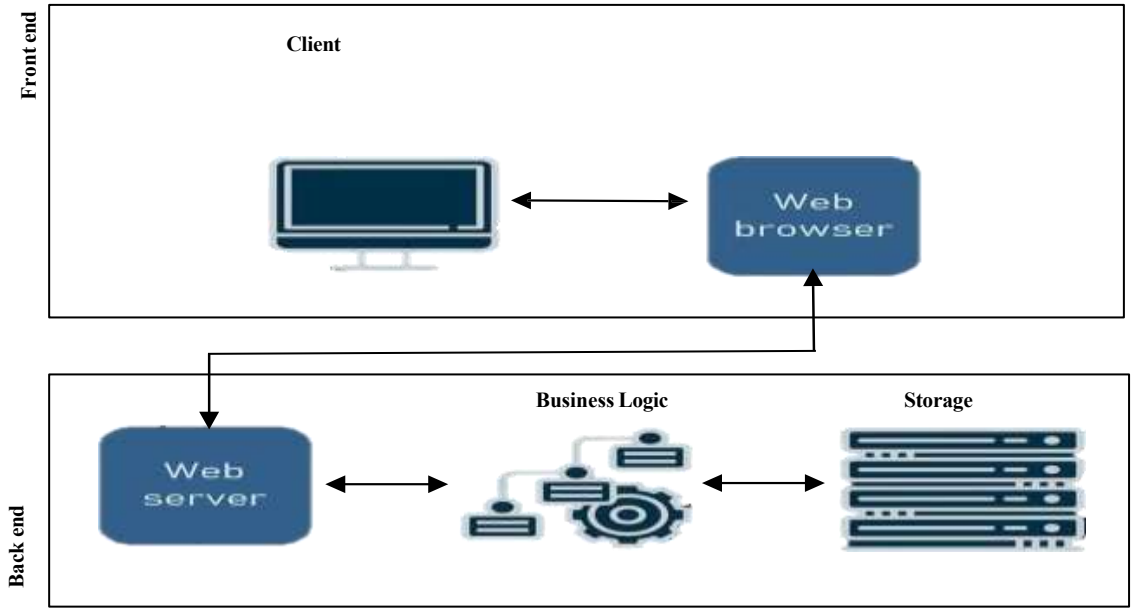


Fig.4.1: System Architecture diagram

The system follows a client–server architecture, where users access the platform through a web browser on the client side. The web server handles incoming requests, processes them, and communicates with the backend components. Business logic modules manage core operations such as donor/recipient registration, blood availability checks, and request processing. All records are securely stored and retrieved from a centralized database server. This architecture ensures smooth interaction, efficient data processing, and reliable storage for real-time blood bank management.

4.2 ENTITY RELATION:

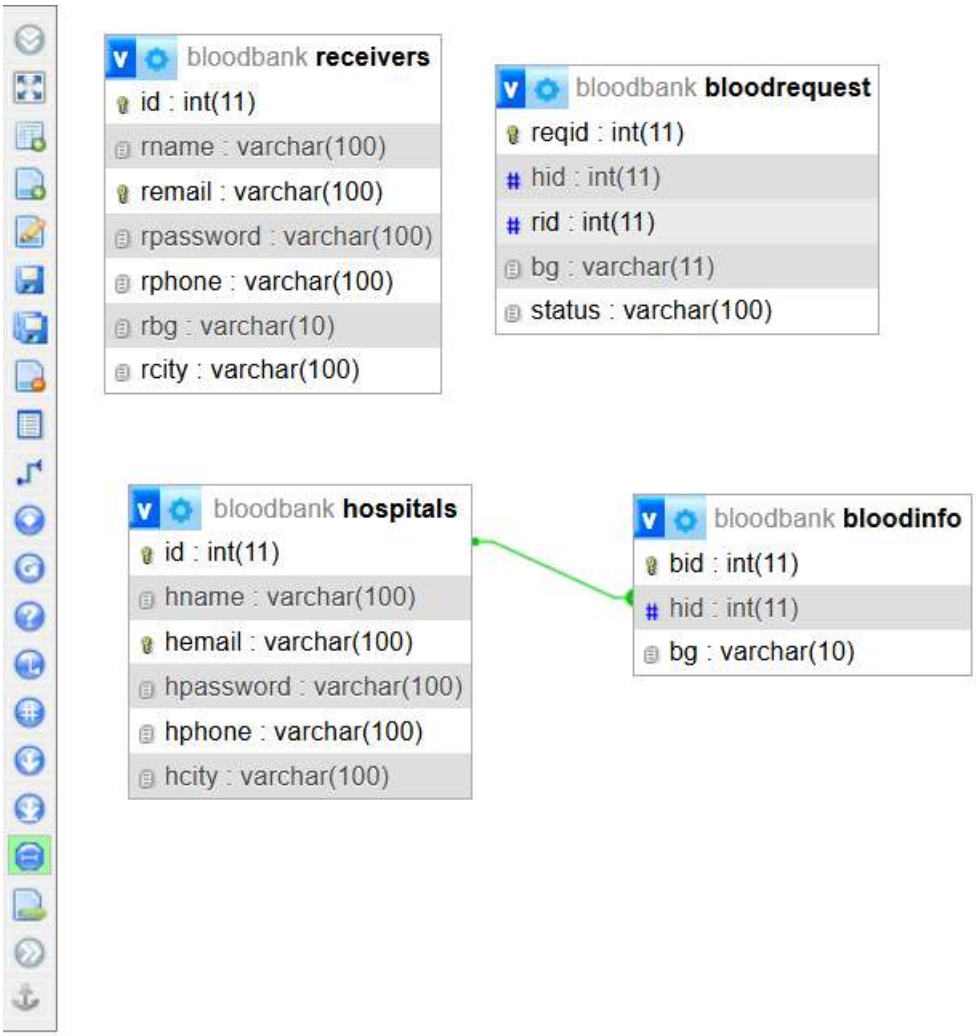


Fig.4.2: Entity Relation Diagram

This database architecture diagram illustrates the structure and relationships among various tables that define the data organization of the system.

4.3 DATA FLOW DIAGRAM:

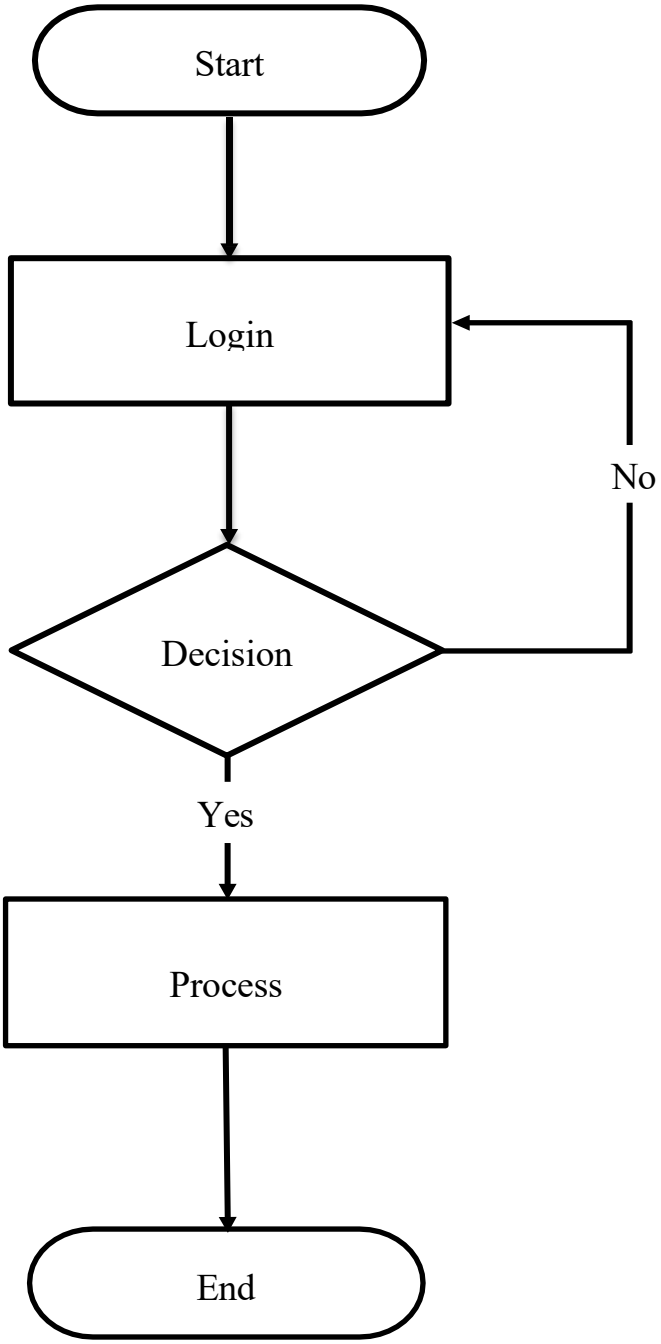


Fig.4.3: Data flow diagram

The flowchart illustrates the system login process, showing the sequence of steps from user authentication to successful process execution.

CHAPTER 5

IMPLEMENTATION

5.1 ALGORITHMS

These algorithms form the backbone of the blood bank system, ensuring accurate data processing, secure access, and real-time monitoring of donor and recipient activities. They collectively enable efficient management of blood records and enhance overall system reliability:

1. User Authentication and Authorization:

Registration Algorithm:

- Collect user details (name, email, password).
- Validate input (e.g., email format, password strength).
- Check if the email already exists in the database.
- Hash the password.
- Store user details in the database.

```
# --- Register ---
 if (isset($_POST['register'])) {
    $n = $_POST['name']; $e = $_POST['email']; $p = $_POST['password']; $r = $_POST['role'];
    if (!filter_var($e, FILTER_VALIDATE_EMAIL)) echo "Invalid email!";
    else {
       $q = $conn->prepare("SELECT id FROM users WHERE email=?");
       $q->bind_param("s", $e); $q->execute(); $q->store_result();
       if ($q->num_rows) echo "Email exists!";
       else {
          $p = password_hash($p, PASSWORD_DEFAULT);
          $s = $conn->prepare("INSERT INTO users(name,email,password,role) VALUES(?,?,?,?)");
          $s->bind_param("ssss", $n, $e, $p, $r);
          echo $s->execute() ? "Registered!" : "Error!";
```

```
if (isset($_POST['login'])) {
$e = $_POST['email']; $p = $_POST['password'];
$s = $conn->prepare("SELECT id,name,password,role FROM users WHERE email=?");
$s->bind_param("s", $e); $s->execute(); $r = $s->get_result();
if ($u = $r->fetch_assoc() and password_verify($p, $u['password'])) {
  $_SESSION = $u;
} else echo "Login failed!";
}
if (isset($_GET['logout'])) { session_destroy(); header("Location:?"); }
if (isset($_SESSION['id'])) {
echo "<h3>Welcome {$_SESSION['name']} ({$_SESSION['role']})</h3>";
echo "<a href='?logout=1'>Logout</a>";
exit;}
```

2. Place Search and Filtering:

Search Algorithm:

- To allow users to search for donors in a specific place.
- To filter donors by blood group or other criteria.
- To provide quick and accurate search results from the donor database.

```
<?php
$c=new mysqli("localhost","root","","blood_bank");
$p=$_POST['place']??""; $b=$_POST['blood']??"";
$r=$c->query("SELECT * FROM donors WHERE place LIKE '%$p%' OR blood_group='$b'");
echo $r->num_rows ? join("<br>", array_map(fn($x)=>"$x[name] | $x[blood_group] | $x[place]", $r->fetch_all(MYSQLI_ASSOC))) : "No donors found."; ?> }
```

3. Recommendation System:

Basic Recommendation Algorithm:

- To recommend the most compatible donors for a recipient.
- To reduce the time needed for manual searching.
- Recommend places based on similar user activities.

```
function recommendDonors($recipientBlood, $place, $pdo) {
// Step 1: Define blood group compatibility
$compatibility = [
  "A+" => ["A+", "A-", "O+", "O-"],
  "B+" => ["B+", "B-", "O+", "O-"],
  "O+" => ["O+", "O-"],
  "AB+" => ["A+", "A-", "B+", "B-", "O+", "O-", "AB+", "AB-"]
];
```

4. Admin Panel Operations:

Add/Update/Delete Place Algorithm:

```
<?php
$conn = new mysqli("localhost", "root", "", "blood_bank");
if ($conn->connect_error) die("DB Connection Failed");
$op = $_POST['action']; // add / update / delete
if ($op == "add") {
$sql = "INSERT INTO places (name, description, location, type, rating, image, video)
    VALUES ('City Blood Bank','Main Center','Delhi','Hospital',4.5,'img.jpg','vid.mp4')";
$conn->query($sql);
echo "Place Added Successfully";
```

```
    }
  if ($op == "update") {
     $sql = "UPDATE places SET name='Metro Blood Bank', location='Mumbai', rating=5 WHERE id=1";
 $conn->query($sql);
 echo "Place Updated Successfully";
}
if ($op == "delete") {
 $sql = "DELETE FROM places WHERE id=2";
 $conn->query($sql);
 echo "Place Deleted Successfully";
}
$conn->close();
?>
```

These algorithms are foundational for A Web-based Blood Bank System for Managing Records of Donors and Receipts. You can expand and optimize them further based on specific requirements and user feedback.

CHAPTER 6

SOFTWARE TESTING

6.1 SCOPE

The testing phase ensures that the Automated Notes Maker from Audio Recordings does function, fulfills its specifications, and provides a highly dependable and friendly experience. It varies from unit testing to system testing to detect and correct any problem or bug found.

6.2 TESTING METHODS

6.2.1 BLACK-BOX TESTING

Black-box testing is a software testing method where the internal logic or code of the system is not examined. Instead, the focus is on testing the system's functionality by providing different inputs and checking whether the actual outputs match the expected results. It ensures that the system behaves correctly from a user's point of view:

Advantages of Black Box Testing:

- **Real-World Usage Focus:** Helps ensure the system works as expected for end users, improving usability and user satisfaction.
- **Better at Finding Functional Errors:** Detects missing functions, incorrect outputs, interface issues, and data handling problems effectively.
- **Independent Testing:** Testing is unbiased because testers are not influenced by the developer's code or logic.

Black Box Testing Techniques for A Web-based Blood Bank Website:

- **Equivalence Partitioning:** Divide website functionalities into categories (e.g., search function, user login, and itinerary creation) and design test cases for valid and invalid inputs within each category.
    - Test cases for search function could include searching by location name, keyword, and category, along with misspelling variations and empty searches.

- **Boundary Value Analysis:** Focus on testing functionality at the edges or boundaries of expected inputs.

    - Test searching locations with very short or very long names, entering extreme values for date filters in itinerary creation, or submitting login forms with minimum and maximum character password lengths.

- **Error Guessing:** This technique involves anticipating potential errors users might encounter and designing test cases to trigger those errors.
    - Test how the website responds to unexpected user inputs like special characters in search queries, submitting incomplete forms, or rapid clicks on navigation menus.
- **User Scenario Testing:** Develop test cases based on realistic user scenarios.
    - Create scenarios like a first-time donor registering to donate blood, a recipient searching for available blood units, or an admin managing donor and recipient records in the Web-based Blood Bank System for Managing Records of Donors and Receipts.
    - Test the website's ability to guide users through these scenarios and fulfill their information needs.
- **Usability Testing:** While usability testing is a separate category, it can be incorporated into black box testing by involving testers from your target audience.
    - Observe how they interact with the website, identify any confusing elements, and assess the overall user experience.

Tools for Black Box Testing:

- Tools like Selenium, QTP (UFT), and Test Complete are used to test the functionality of the Blood Bank System — such as donor registration, login, and search operations — without checking the internal code.
- Helps test user interfaces such as input forms, search options, and navigation links in the Blood Bank System.

Benefits of Black Box Testing in Your Project:

- Identify issues impacting user experience and overall website effectiveness.
- Enhance the website's usability and make it more user-friendly for your target audience.
- Ensure the website functions as intended from a user's perspective.
- Contribute to a successful launch of your a web-based blood bank website.

Remember, black box testing is most effective when combined with other testing methods like performance testing and security testing. By employing a comprehensive testing strategy, you can ensure your A Web-based Blood Bank website offers a reliable and enjoyable experience for users.

6.2.2 WHITE-BOX TESTING

White box testing, also known as glass box testing or code-based testing, is a software testing methodology where the internal structure and code of the website are examined. This approach leverages knowledge of the code to design test cases that thoroughly explore specific functionalities and identify potential issues within the website's logic.

While black box testing focuses on user experience, white box testing delves deeper to ensure the website functions as intended based on its underlying code. Here's how white box testing can benefit you're a web-based blood bank website project:

Advantages of White Box Testing:

- **Targeted Testing:** By examining the code, testers can design specific test cases to target critical functionalities and identify code-related errors. This allows for in-depth testing of complex website features.
- **Improved Code Coverage:** White box testing helps ensure a high percentage of the website's code is executed during testing, minimizing the risk of untested code causing issues later.
- **Logic Error Detection:** This method is adept at uncovering logic errors within the code that might not be apparent through user-centric testing.

White Box Testing Techniques for A Web-based Blood Bank Website:

- **Statement Coverage:** This technique ensures every line of code is executed at least once during testing. This is a foundational approach to ensure all functionalities are at least triggered.
- **Branch Coverage:** This method focuses on testing all possible execution paths within the code. This involves testing different conditions (if/else statements) and ensuring the code behaves as expected for each path.
- **Data Flow Testing:** This technique maps the flow of data through the website's code. Test cases are designed to ensure data is processed accurately and manipulated according to intended logic.
- **Loop Testing:** Thoroughly tests various loop functionalities within the code. This includes testing loops with different iteration counts, boundary values, and potential infinite loop scenarios.

Who Performs White Box Testing?

White box testing typically requires a strong understanding of the website's programming language and code structure. Testers with web development experience or access to developers familiar with the website's codebase are best suited for this type of testing.

Considerations for White Box Testing:

- **Complexity:** White box testing can be time-consuming and complex, especially for large websites with intricate codebases.
- **Resource Dependence:** The effectiveness of this method relies heavily on the expertise and understanding of the testers involved.
- **Focus:** While valuable, white box testing shouldn't replace black box testing. It's most effective when used in conjunction with user-centric testing approaches.

Benefits of White Box Testing in Your Project:

- Ensures the website's code functions as intended and adheres to its designed logic.
- Identifies hidden errors within the code that might impact functionality.
- Improves the overall code quality and reliability of your a web-based blood bank website.

By combining white box testing with black box testing methods, you can achieve comprehensive testing coverage for your a web-based blood bank website. This dual approach guarantees a user-friendly experience while ensuring the website functions flawlessly from a code perspective, leading to a robust and successful launch.

6.2 LEVELS OF TESTING

The testing process for your A Web-based Blood Bank website can be effectively structured across three key levels:

6.2.1 UNIT TESTING

- **Focus:** Individual software units (modules, functions, classes) are tested in isolation to verify they function correctly according to their specifications.
- **Techniques:** Developers typically perform unit testing using automated testing frameworks. Common techniques include testing various input scenarios, expected outputs, and error handling.

- **Benefits for A Web-based Blood Bank Website:**
    - Ensures core functionalities like search algorithms, data manipulation within modules, and user login processes work as intended.
    - Catches bugs early in the development stage, reducing overall testing time and effort.

6.2.2 INTIGRATION TESTING

- **Focus:** Tests how different units or modules interact and function together as a system.
- **Process:** After successful unit testing, individual units are integrated and tested to ensure they communicate seamlessly and exchange data correctly.
- **Techniques:**
    - Developers or testers might use a combination of manual and automated testing to simulate interactions between modules.
    - This could involve testing user login integration with database access or interaction between search functions and map displays.
- **Benefits for A Web-based Blood Bank Website:**
    - Identifies issues arising from communication gaps between different website components.
    - Ensures data flows smoothly between modules and functionalities work cohesively.

6.2.3 SYSTEM TESTING

- **Focus:** Evaluates the entire website as a complete system to ensure it meets all functional and non-functional requirements.
- **Scope:**
  - System testing encompasses all functionalities, user experience, performance, security, and compatibility across different browsers and devices.
  - This level incorporates black box testing techniques to simulate real-world user scenarios.
- **Techniques:**
  - A combination of manual and automated testing approaches is used, involving user acceptance testing (UAT) with representatives from your target audience.
  - Performance testing tools can evaluate website loading speed and stability under various user loads.
- **Benefits for A Web-based Blood Bank Website:**
  - Identifies critical issues impacting overall website functionality and user experience.
  - Ensures the website performs well under real-world conditions and is ready for launch.

Testing Levels Working Together:

The mini-project **"A Web-based Blood Bank System for Managing Records of Donors and Recipients"** is designed to provide a digital platform for organizing and managing blood donation activities efficiently. The system allows donors to register with their personal and medical details, while recipients can request blood by specifying the required blood group and contact information. It helps hospitals and administrators maintain accurate records, verify donor details, check blood availability, and respond to emergency requests quickly.

The platform is intended for medical and humanitarian purposes only, and users are expected to provide accurate details and follow ethical usage guidelines. While the system stores and organizes donor information, medical fitness for blood donation must still be verified by certified health professionals. The system aims to improve response time in emergency situations, support hospitals in managing blood inventory, and promote voluntary blood donation. Overall, this web-based solution improves efficiency and transparency in blood management and contributes to saving lives by connecting donors and recipients faster and more effectively.

CHAPTER 7

RESULT AND SNAPSHOTS

Receiver Interface:

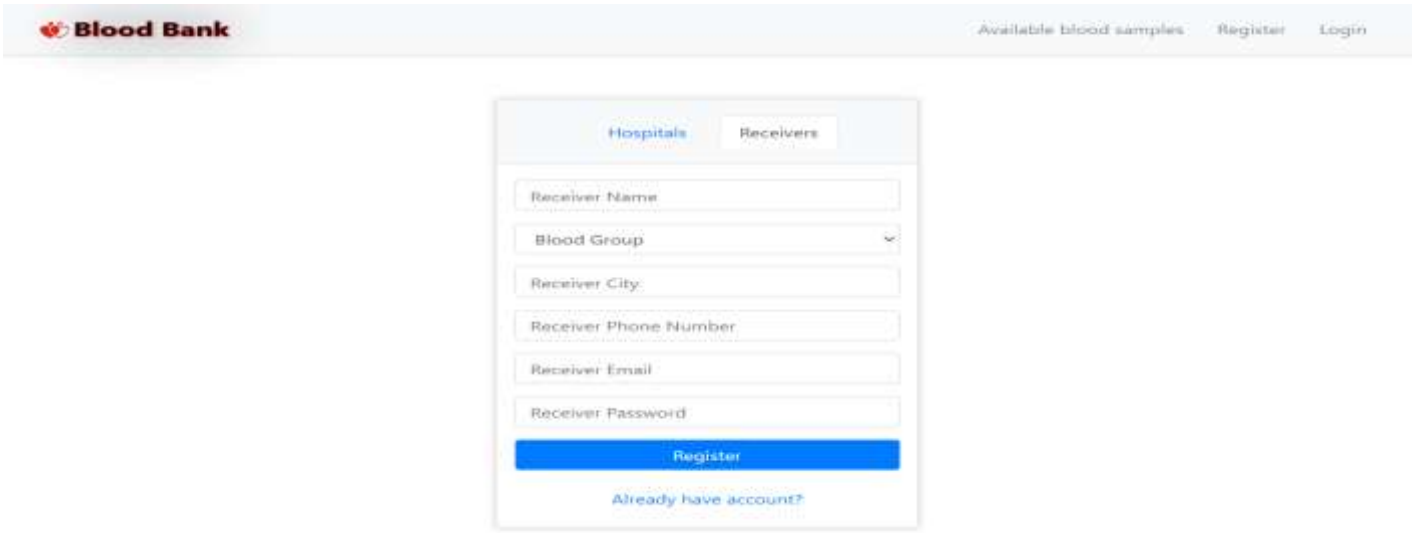
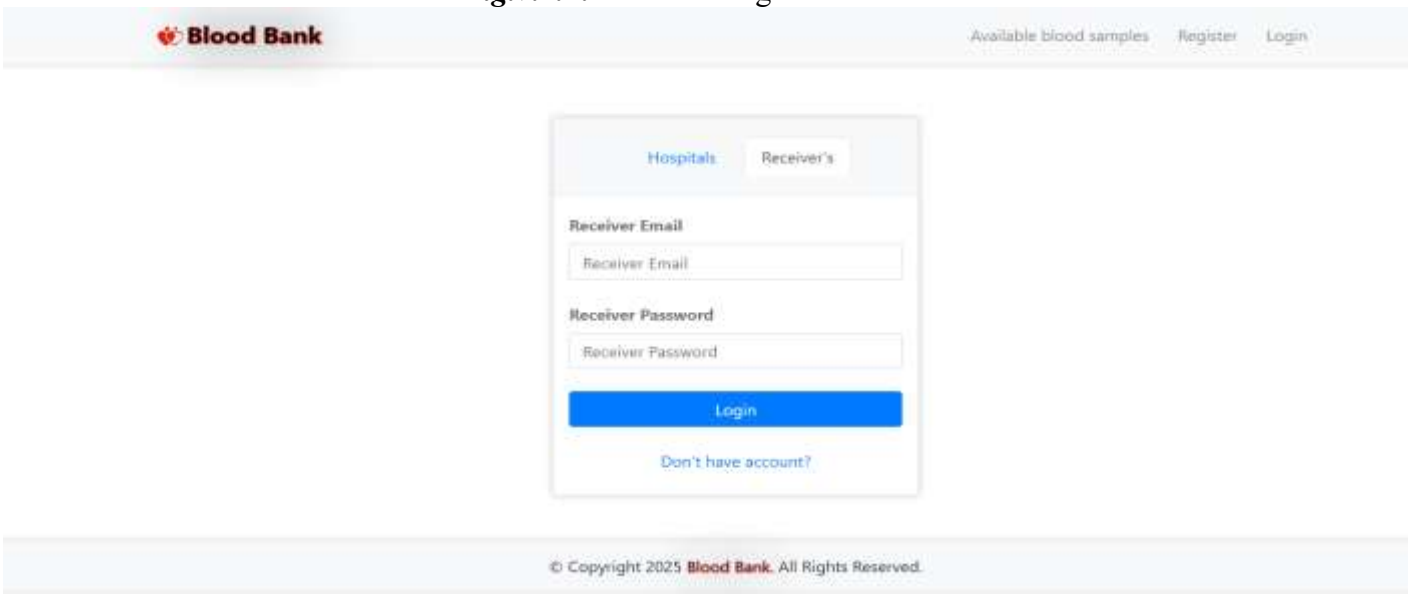


Fig.7.1.1: Receiver Register

This form represents the receiver registration module where users can input their details to register in the system.

Fig.7.1.2: Receiver Login



This diagram represents the receiver login module where users can authenticate themselves to access the system.

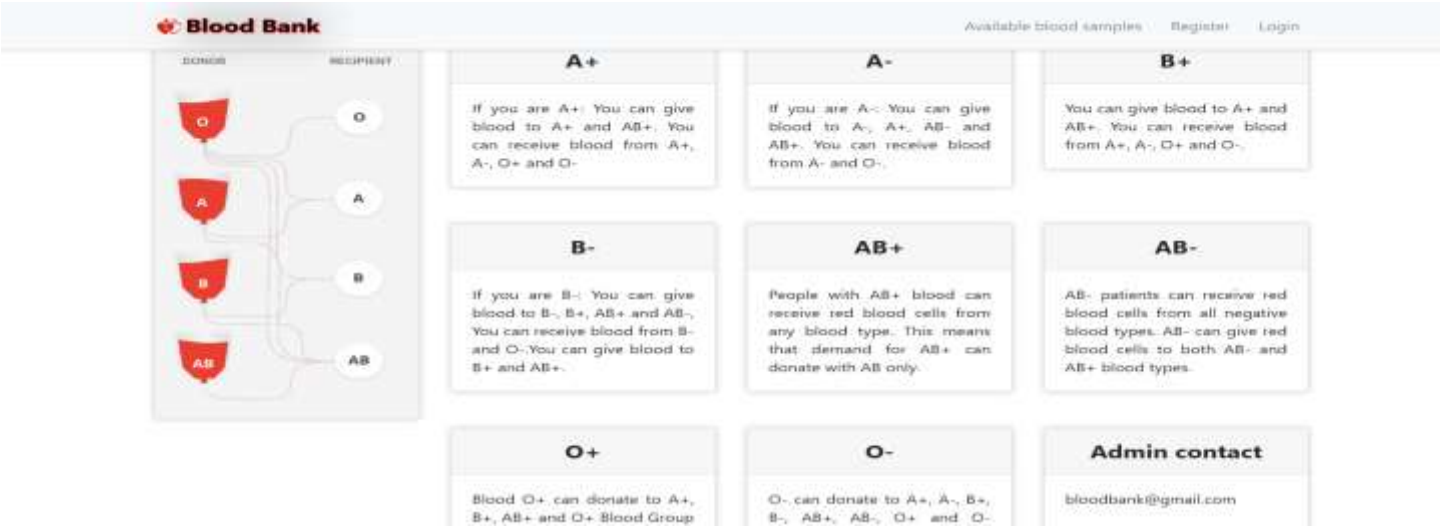


Fig.7.1.3: User Home Page

This interface represents the user home page showing available blood group compatibility details and admin contact information.



Fig.7.1.4: Choose Blood Bank

The interface displays the list of blood bank requests showing details such as name, contact information, blood group, status, and available actions.

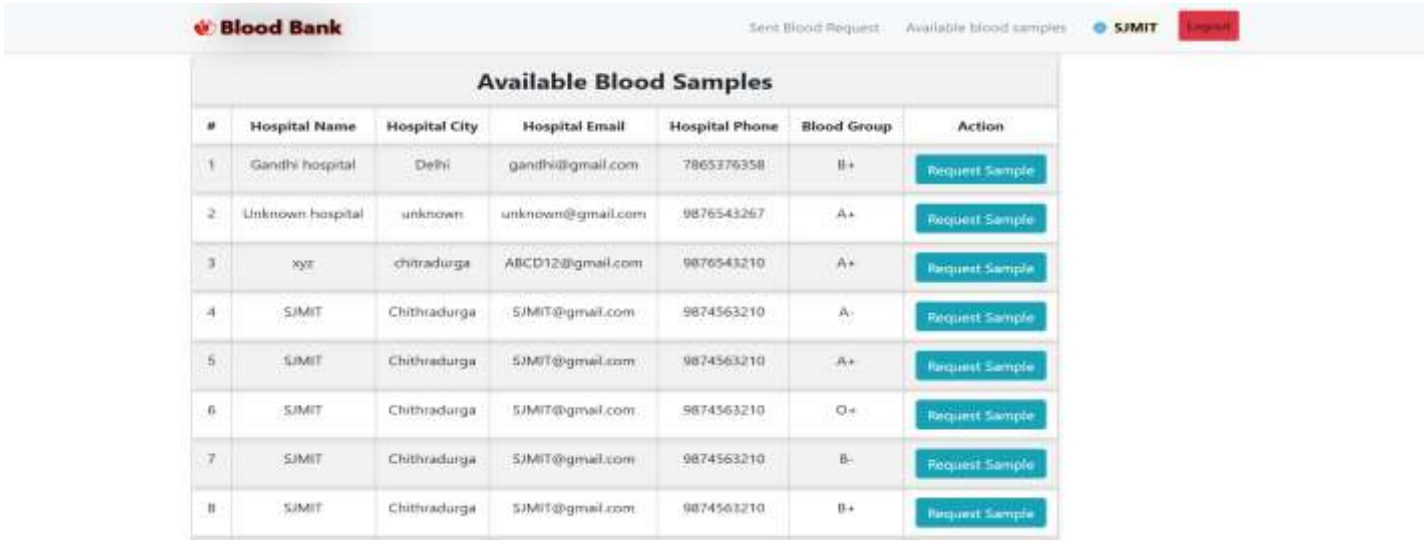


Fig.7.1.5: Available Blood Samples

The interface displays a list of available blood samples along with hospital details, contact information, and an option to request specific samples.

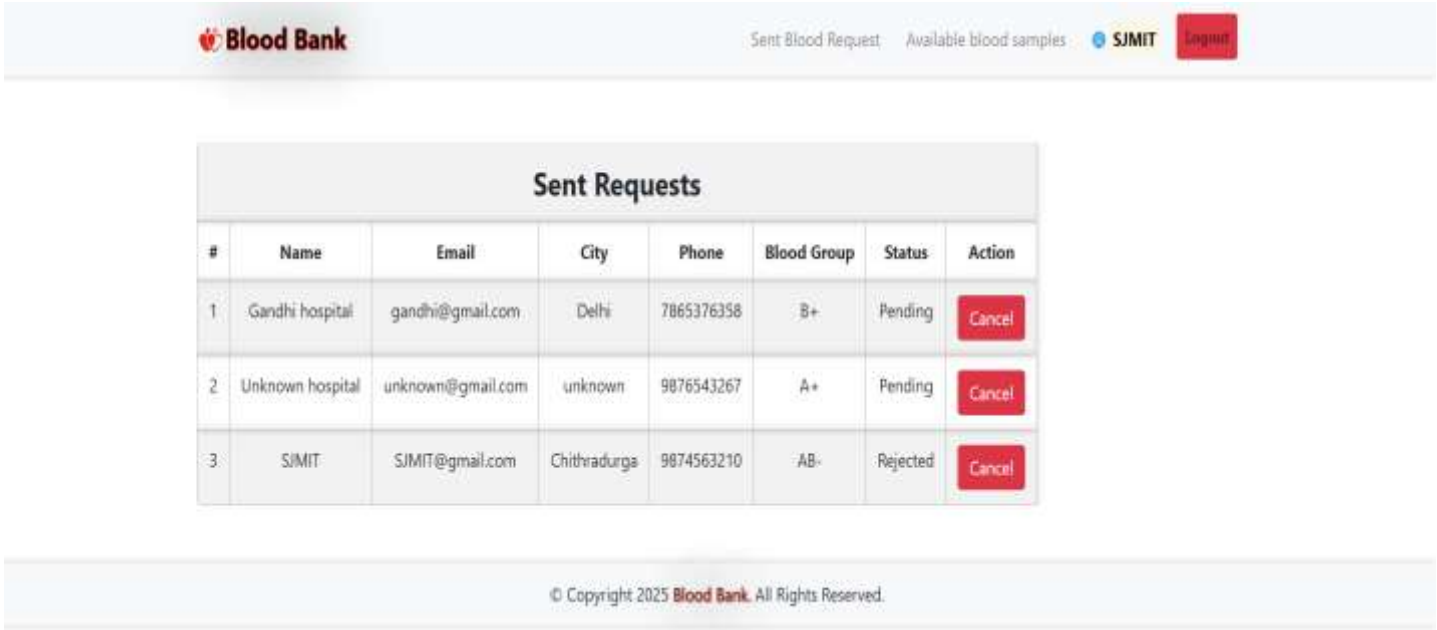


Fig .7.1.6: Sent requests

The interface displays the sent blood requests with details such as hospital name, contact information, blood group, current status, and cancel option.

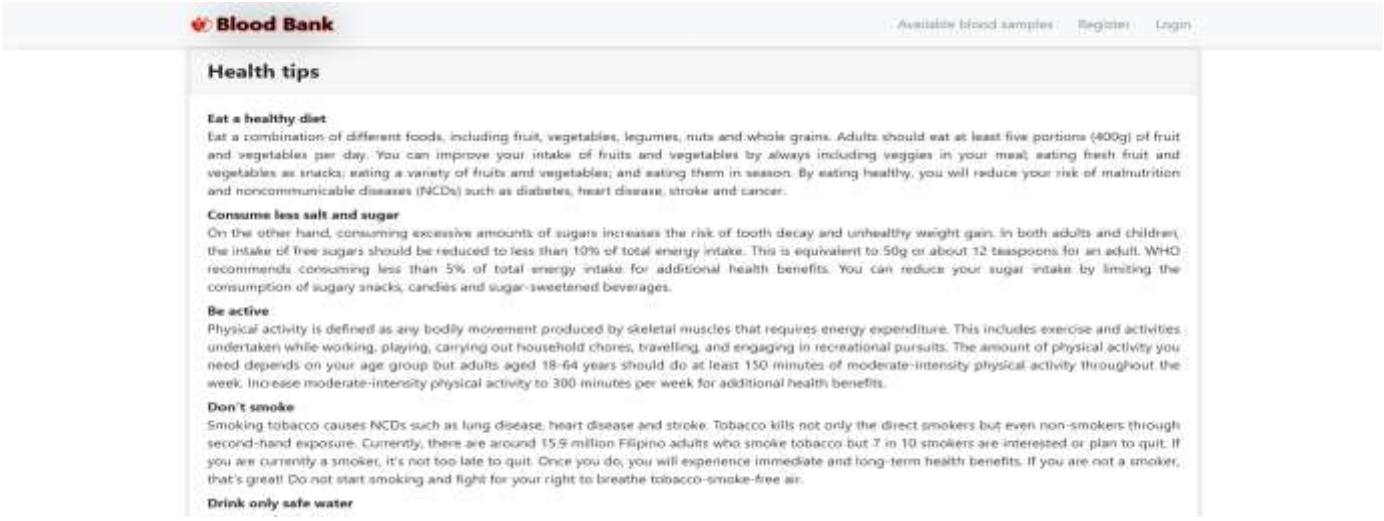


Fig.7.1.7: Health tips

The page provides essential health tips promoting a balanced diet, physical activity, and healthy lifestyle habits to enhance overall well-being.

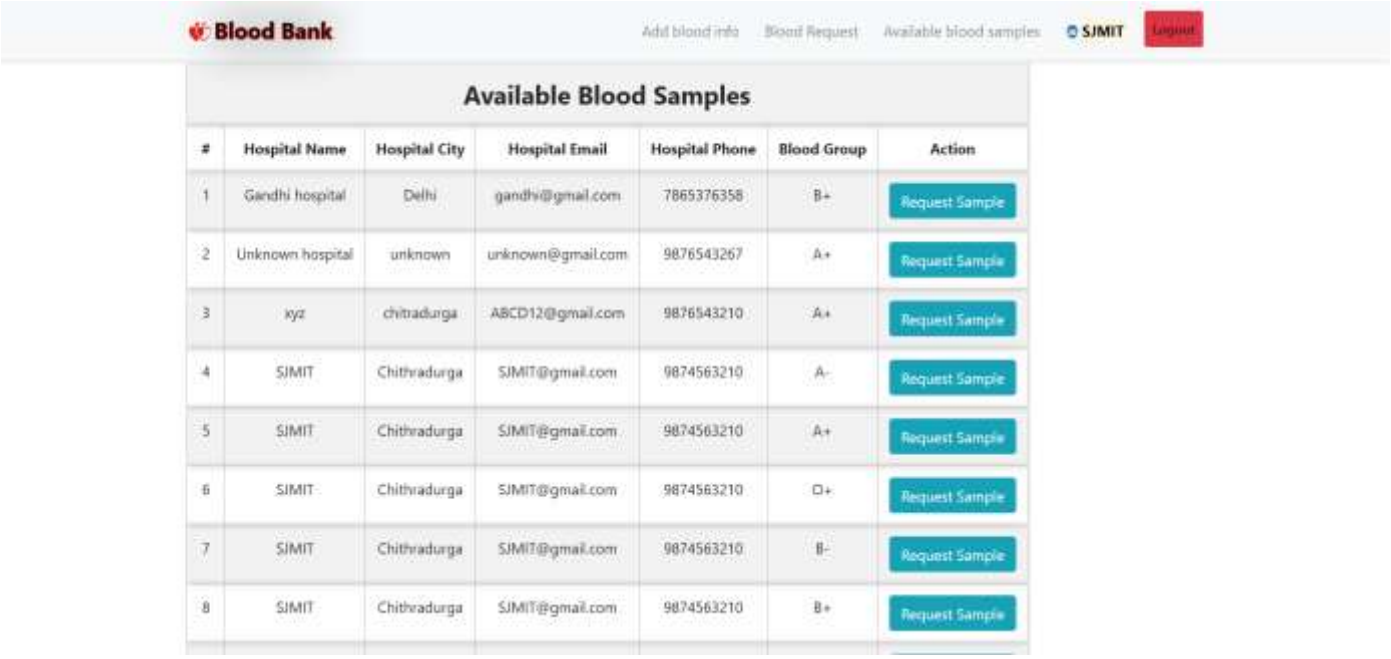


Fig.7.1.8: *Available Blood Samples*

The interface displays a list of available blood samples along with corresponding hospital details, contact information, and an option to request samples.



Fig.7.1.9: *Blood Type Compatibility*

The diagram illustrates blood type compatibility, showing which blood groups can donate to or receive from others along with contact details for assistance.

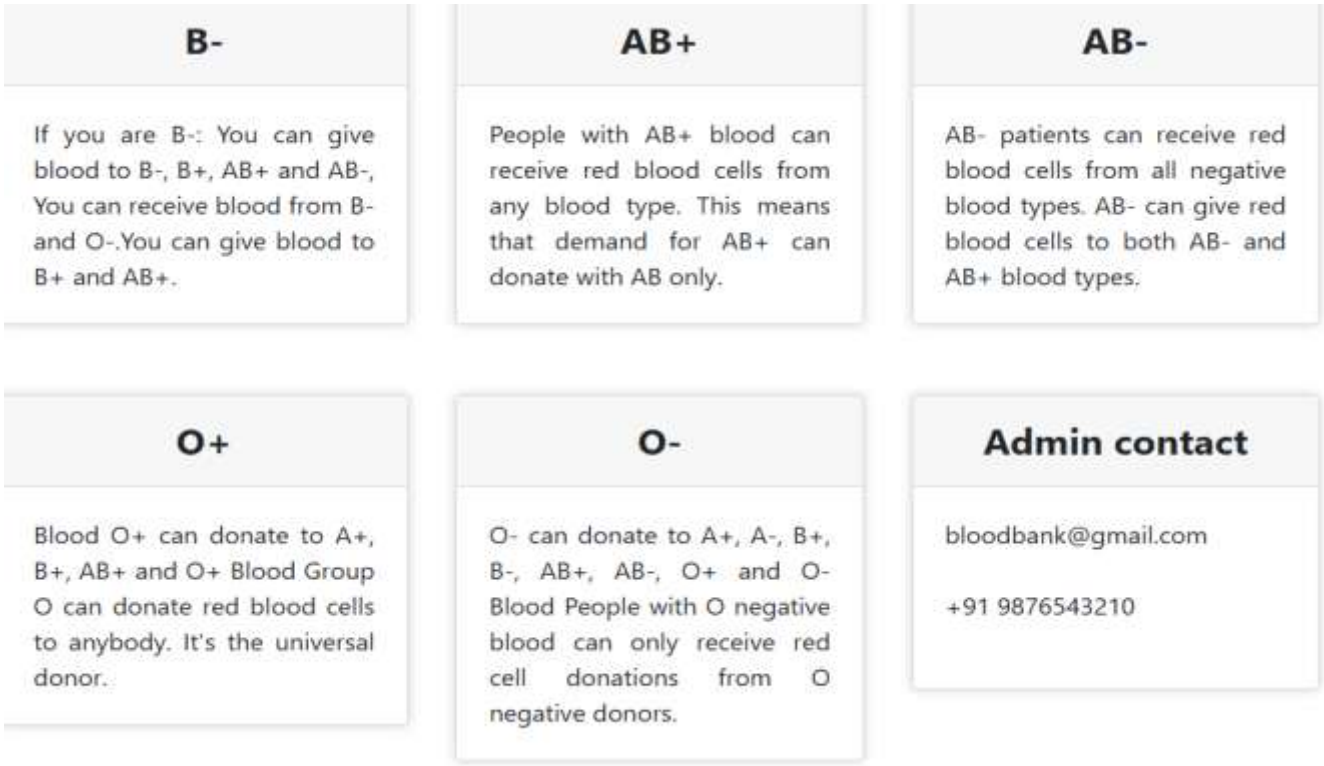


Fig.7.1.10: *Admin contact*

Provides details of blood group compatibility along with contact information for blood donation assistance.

7.1 Hospital Interface:

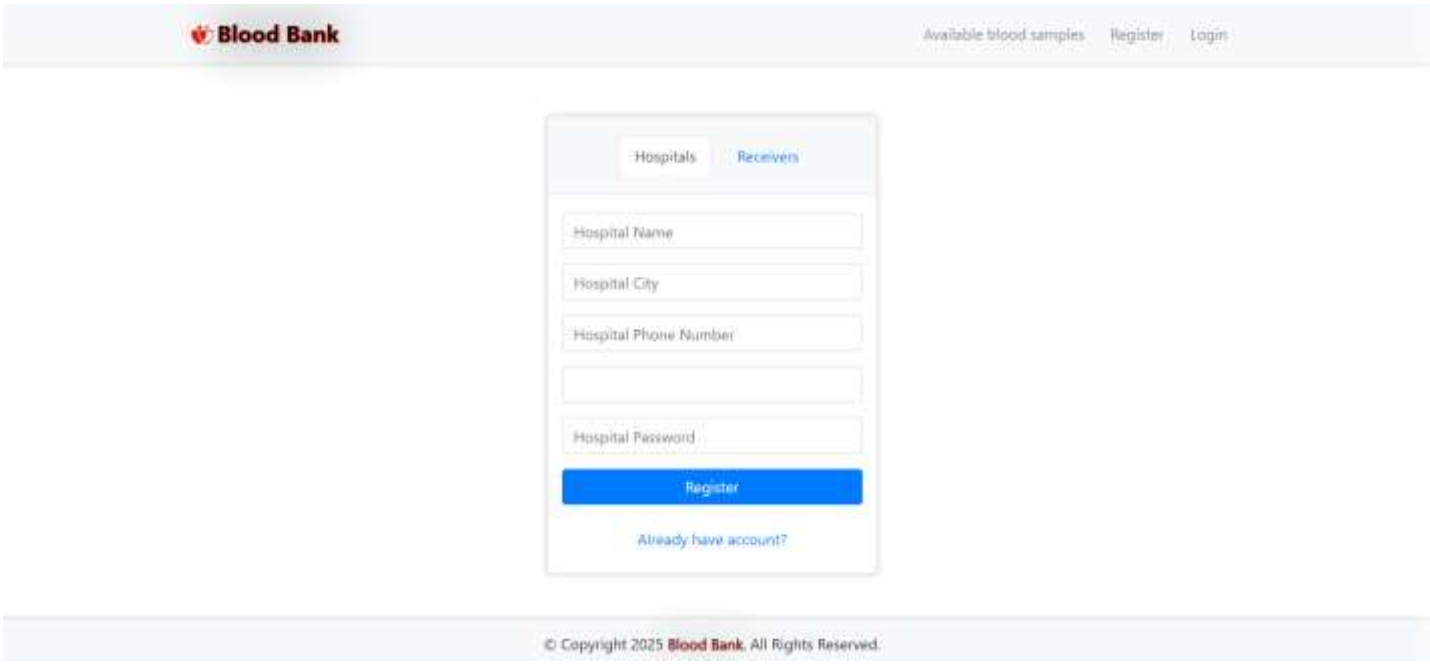


Fig.7.2.1: Hospital Login

This section shows the hospital login interface, allowing hospitals to register or log in by providing essential details such as name, city, phone number, and password.

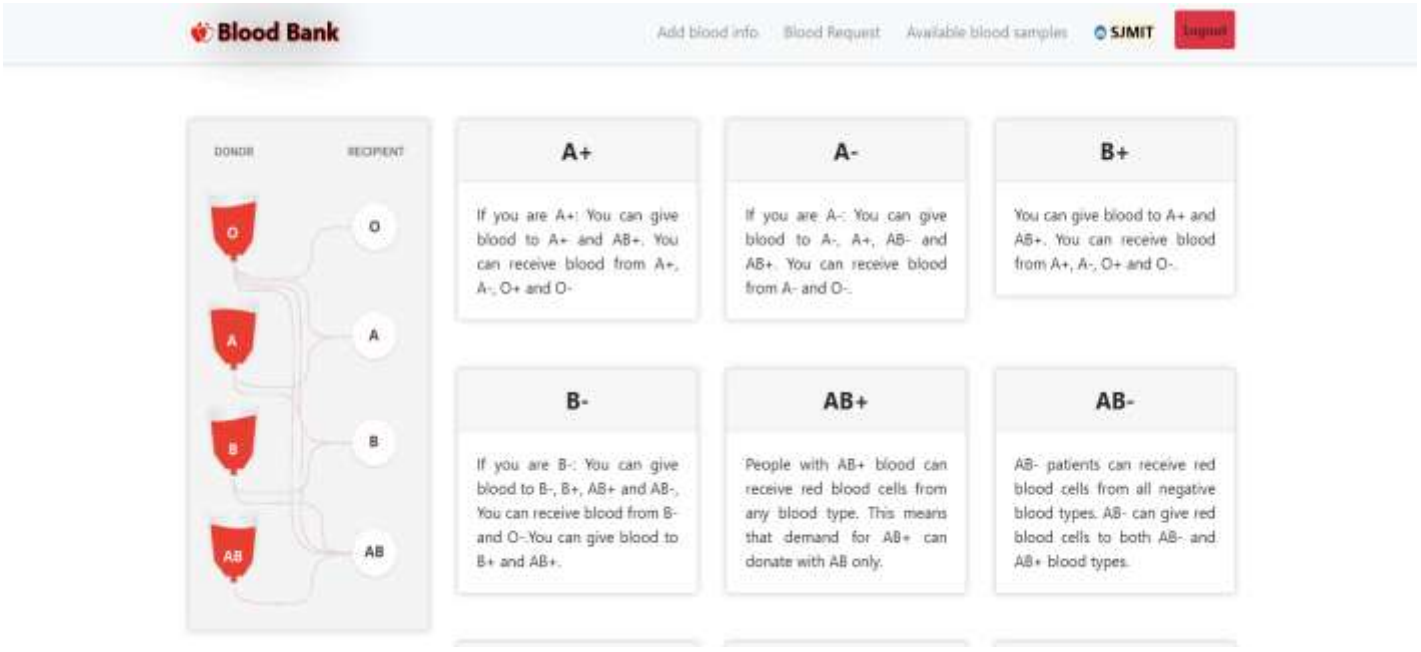


Fig.7.2.2: Hospital Home Page

This page displays detailed blood type compatibility information, helping hospitals identify suitable donors and recipients for safe blood transfusions.

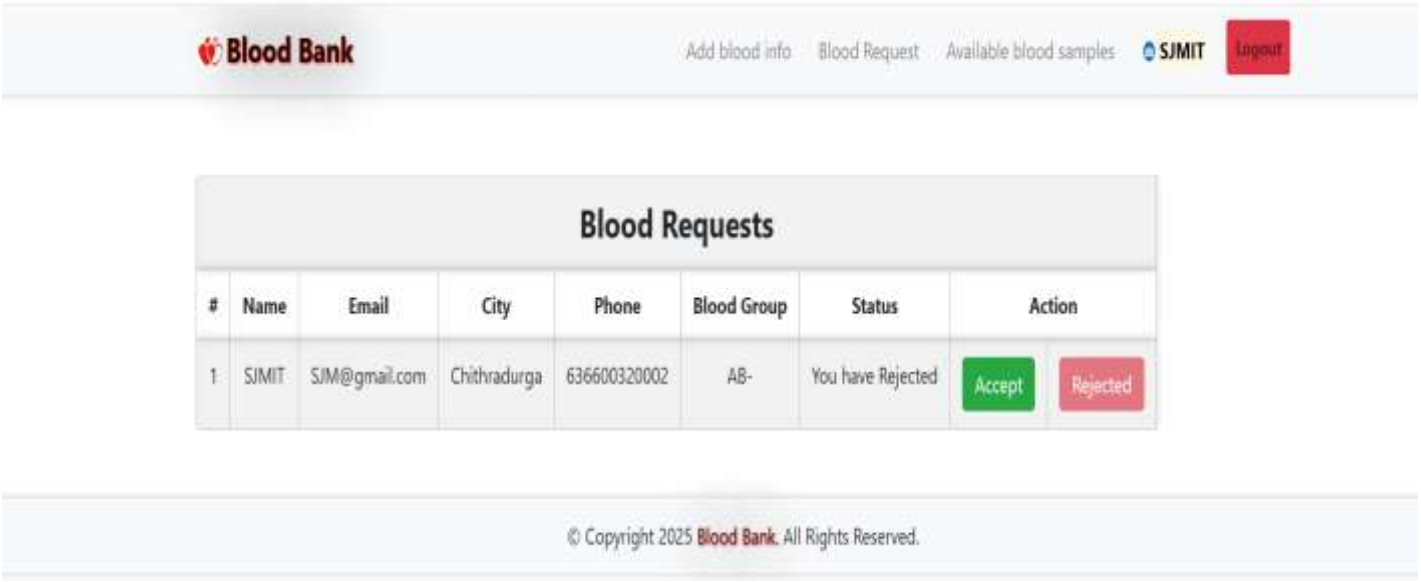


Fig.7.2.3: Blood requests

This page displays blood request details including requester information, blood group, and request status, allowing hospitals to accept or reject blood donation requests efficiently.

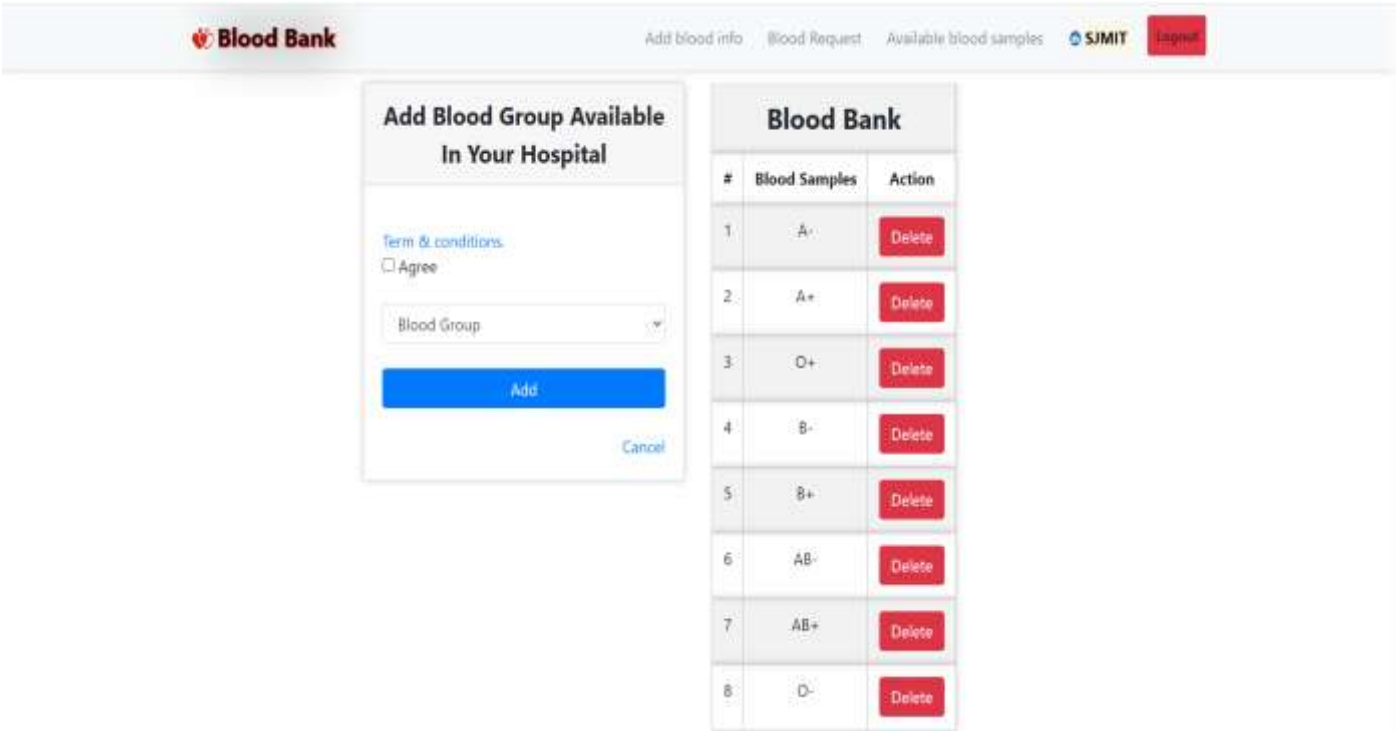


Fig.7.2.4: Blood Bank

This page allows hospitals to manage their available blood samples by adding new blood groups or deleting existing ones from the blood bank record.

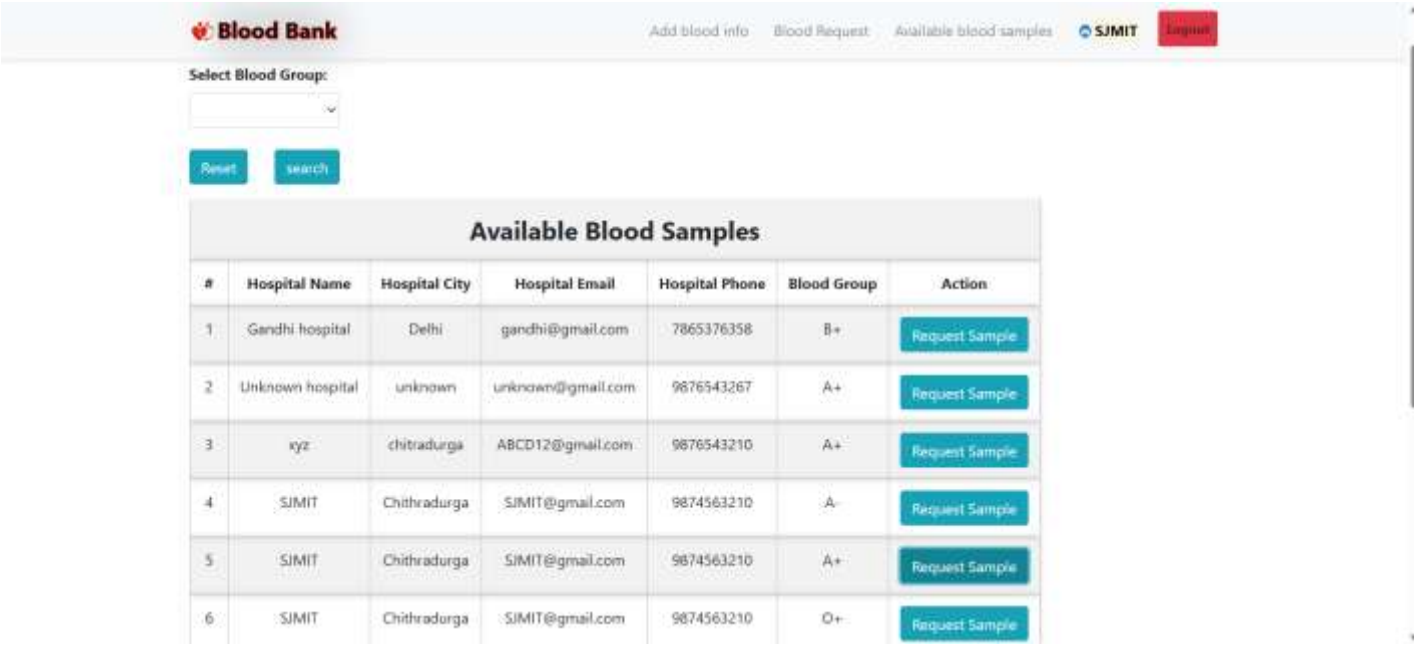


Fig.7.2.5: Available Blood Samples Hospital

This page displays a list of hospitals with their available blood samples, allowing users to search by blood group and request samples as needed.

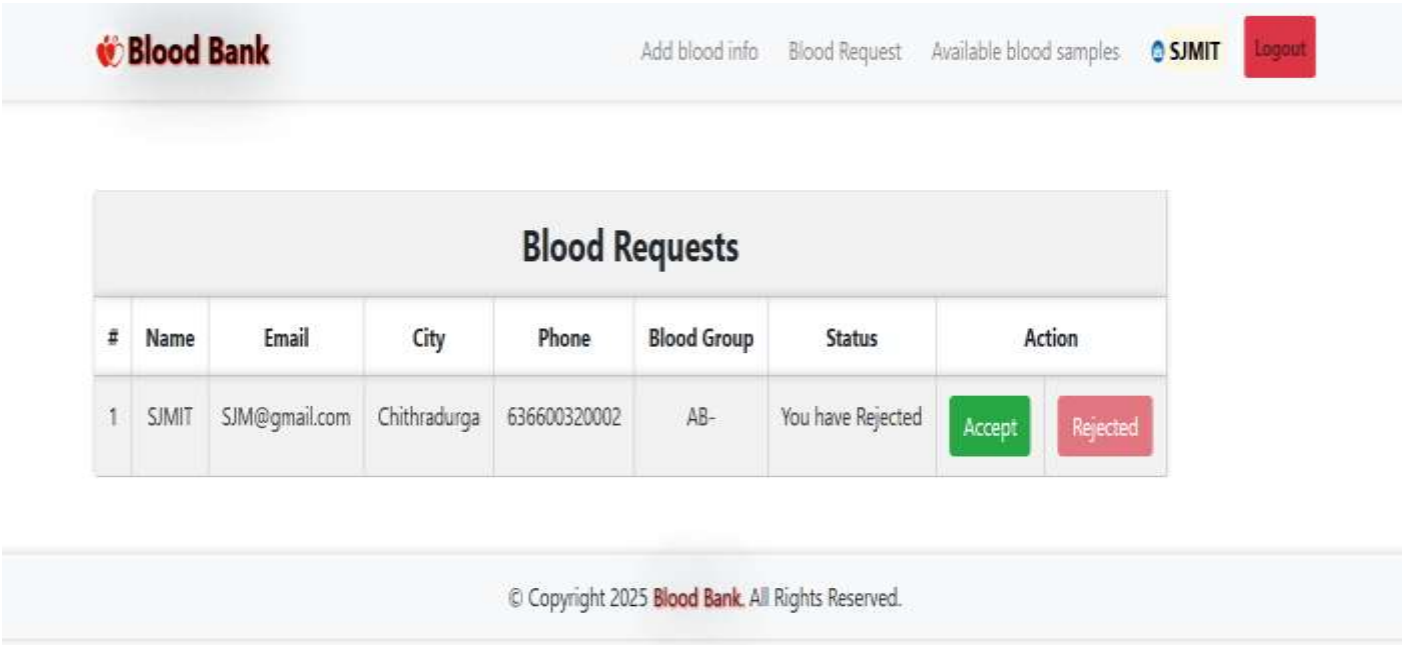


Fig.7.2.6: Blood requests Status

This page displays the status of blood requests, allowing hospitals to view, accept, or reject donor requests efficiently.

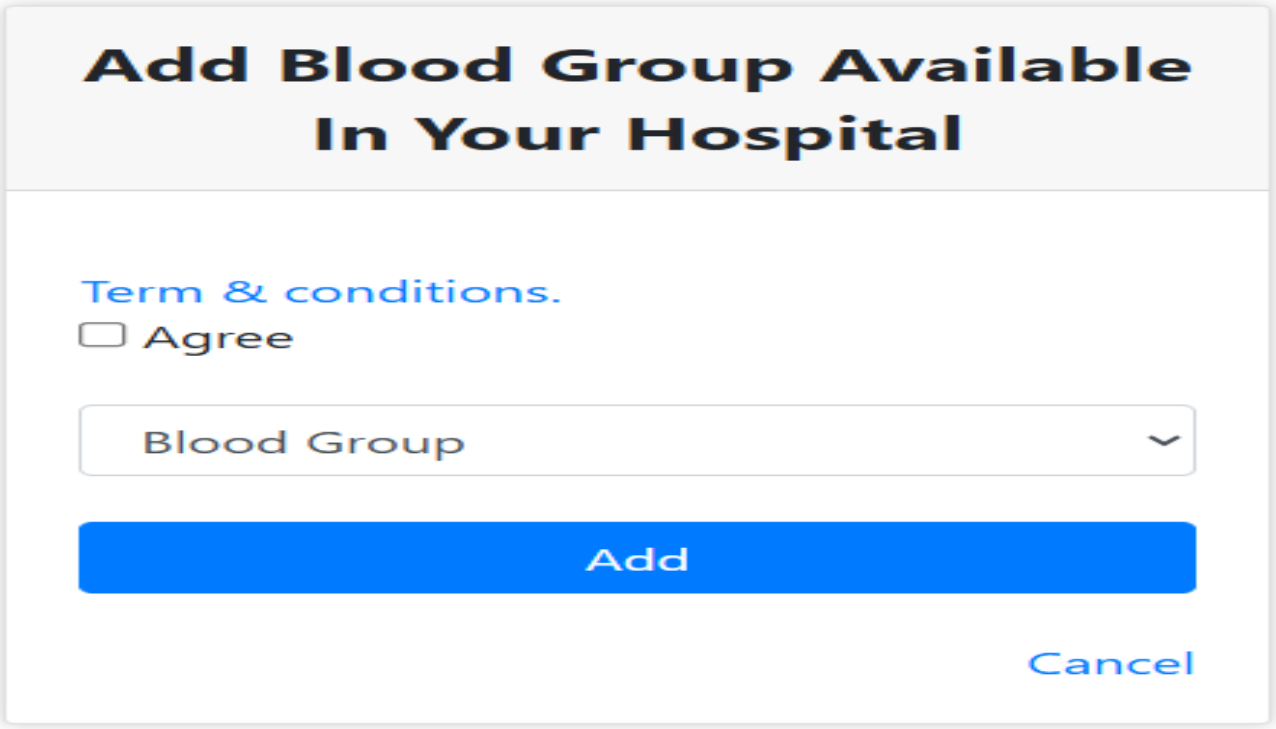


Fig.7.2.7: Add blood group available in your hospital

This page allows hospitals to add available blood groups to their database by selecting the type and agreeing to the terms and conditions.

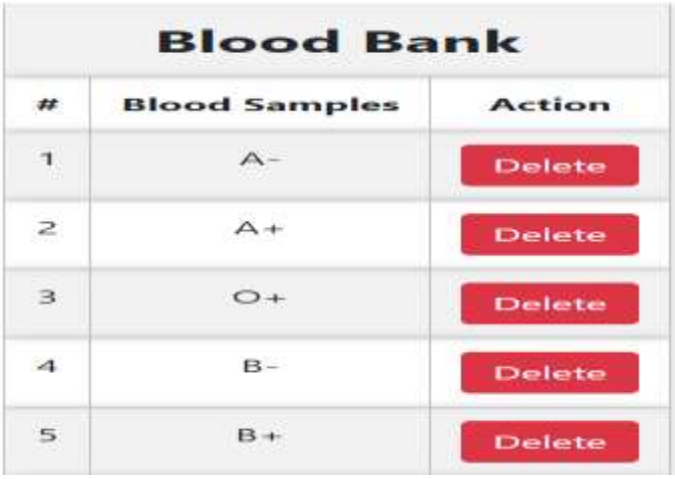


Fig .7.2.8: *Blood Samples*

This section displays the list of available blood samples in the blood bank, allowing hospitals to manage and delete records as needed

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion

The Web-based Blood Bank System for Managing Records of Donors and Receipts provides an efficient, reliable, and user-friendly platform for managing blood donation activities. It simplifies the process of maintaining donor and recipient information, reduces manual work, and ensures quick access to blood availability in emergencies. By connecting hospitals, donors, and blood banks through a centralized online system, it promotes timely donations and saves lives. Overall, this system enhances the efficiency of blood management and contributes significantly to better healthcare services.

Future Enhancements

The Web-based Blood Bank System for Managing Records of Donors and Receipts is designed to simplify and automate the process of managing blood donations. The system connects donors, recipients, hospitals, and blood banks through a centralized online platform, ensuring that blood availability information is accurate and easily accessible. It minimizes manual work, reduces errors, and improves response time during emergencies. With features like real-time data updates, secure record handling, and easy search options, the system enhances the efficiency and reliability of blood management. In the future, additional improvements such as mobile app integration, AI-based donor matching, and data analytics can further expand the system's capabilities. Overall, this project contributes to saving lives and improving the healthcare system by promoting timely and organized blood donation processes.

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4. Kayode, A. A., Adeniyi, A. E., Ogundokun, R. O., & Ochigbo, S. A. (2019). An Android based blood bank information retrieval system. Journal of Blood Medicine, 10, 119.


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